

MINERVA / MYRRHA P&ID Symbol Library — Legend Analysis

Authoritative symbol reference for the ongoing MINERVA P&ID reconstruction work, extracted from the client/company standard legend sheet.

Field	Value
Source document	AD_01.16 SUP - PID General Legend Sheet.pdf
Drawing number	106889-PID00 (P&IDs 106889-PID01, sheets 1-9)
Doc number	MYR100PTF-0521
Owner	SCK CEN — MYRRHA / MINERVA project
Revision	A (Released), 16/11/2023
Sheet size / count	A1 · 9 legend sheets (10 PDF pages incl. cover)
Drawn by / Verified / Approved	Leysen J. / Bergmans G. / Voorderhake S.

Standards basis

The legend explicitly subordinates itself to published standards — capture these deviations exactly, because **this company legend is the authoritative source** and takes precedence over generic ISA-5.1 where they differ:

- **ANSI/ISA-5.1-2022** — instrument letter codes (subordinate to Table 4.1) and instrument symbolic representation (subordinate to Table 5.1.1).
- **ISO 10628** — process diagram house style (piping / equipment layout).
- **IEC 60617 (publication 617)** — electrical symbols (Sheet 7). Symbols marked with an asterisk * are **non-standard** (house additions). Relay function numbers marked ** follow **ANSI C37.2**.
- **Tag numbering** — Primary Systems Naming Convention & Terminology SCK CEN/36557490; deviceclass mnemonics SCK CEN/36793249.

Deliverables (this folder)

File	Purpose
<code>legend_symbols.json</code>	Structured catalogue: every symbol + metadata, naming conventions, ISA letter code, bubble matrix, colour/stroke spec, usage notes, and the list of drawn SVG ids.
<code>symbol_library.svg</code>	Reusable SVG <code><defs></code> of 159 <code><symbol></code> elements (zero-size, position:absolute). Reference any symbol with <code><use href="#<id>"/></code> .
<code>symbol_library_preview.svg</code> / <code>.png</code>	Human-readable contact sheet of every drawn symbol with its id (visual QA).
<code>build_legend_library.py</code>	Pure-stdlib generator that (re)produces all of the above. Re-run with <code>python3 build_legend_library.py</code> .

159 symbols are drawn (schematic, standards-based geometry — not a pixel copy of the CAD source) covering the complete process-P&ID scope of sheets 1-5.

The facility-specific (RIB line, beam optics) and electrical/fire sheets (6-9) are catalogued at reference level in `reference_only_sets` (77 listed items).

How to use the SVG library

```

<!-- 1. embed/link symbol_library.svg once (it is defs-only, renders nothing) -->
<!-- 2. place any symbol, scaling freely; each symbol has its own viewBox -->
<svg width="900" height="400">
  <use href="#valve-gate"      x="100" y="100" width="60" height="36"/>
  <use href="#inst-A1"       x="220" y="90"  width="48" height="48"/>
  <use href="#act-diaphragm"  x="320" y="70"  width="70" height="70"/>
  <use href="#line-primary"   x="420" y="110" width="120" height="24"/>
</svg>

```

All ids are stable, lower-case, hyphen-separated, and namespaced by category prefix (`line-` , `inst-` , `flow-` , `valve-` , `act-` , `eq-` , `conn-` , `iface-` , `scope-` , `misc-` , `hvac-`). See `svg_library.drawn_ids` in the JSON for the full list.

1. Line types (Sheet 1)

Line weight is significant and must be honoured:

Symbol id	Name	Line weight	Style
line-primary	Primary line segment	1 mm	solid + arrow
line-secondary	Secondary line segment	0.5 mm	solid + arrow
line-primary-future	Primary, future extension	1 mm	dash-dot + arrow
line-secondary-future	Secondary, future extension	0.5 mm	dash-dot + arrow
line-electrical-signal	Electrical signal	0.25 mm	dashed
line-pneumatic-signal	Pneumatic signal	0.25 mm	double cross-hatch ticks
line-hydraulic-signal	Hydraulic signal	0.25 mm	periodic L-marks
line-software-signal	Software / data signal	0.25 mm	ring markers
line-em-sonic-signal	Guided EM / sonic signal	—	wave marker
line-capillary	Capillary tube	0.25 mm	x cross marks
line-hose	Hose	—	continuous wavy
line-pipe-insulated	Pipe, insulated	—	hatch band
line-tracer	Tracer (heating/cooling)	—	parallel dashed trace
line-jacketed	Jacketed pipeline	—	outer jacket box
line-heated-insulated	Heated/cooled + insulated	—	hatch + trace
line-hvac-supply	HVAC air supply	1 mm	solid
line-hvac-return	HVAC air return	1 mm	dash-dot
conn-connected-lines	Connected lines	—	junction bridge
	Non-connected lines	—	hop / gap

Symbol id	Name	Line weight	Style
conn-non-connected-lines			

2. Instrumentation (Sheet 2 — ISA-5.1 matrix; Sheet 3 — flow elements)

Bubble matrix (type × location) — inst-<TYPE><ROW>

The legend reproduces ISA-5.1 Table 5.1.1 as a 4-column × 5-row matrix. Type is the **enclosure shape**; location is the **line through the bubble**.

Type columns

Col	Meaning	Shape
A	Shared display/control — Basic Process Control System (BPCS)	square enclosing a circle
B	Shared display/control — Safety Instrumented System (SIS)	square + circle + diagonals
C	Computer systems and software	hexagon
D	Discrete visualisation of instruments	plain circle (stadium/oval = two instruments → inst-discrete-oval)

Location rows (accessibility line across the bubble)

Row	Location	Line
1	Field-mounted; operator accessible	none
2	Front of central/main panel or console	single solid
3	Rear of central panel / cabinet behind panel	single dashed
4	Front of secondary/local panel	double solid
5	Rear of secondary/local panel / field cabinet	double dashed

So e.g. `inst-A1` = field BPCS bubble, `inst-B2` = main-panel SIS,
`inst-C3` = behind-panel software, `inst-D5` = field-cabinet discrete.

Instrument tag composition (Sheet 1)

Instrument tags are **not drawn explicitly**; they are composed:

1. take the **first line** of the tag of the element the instrument is attached to;
2. append the **top + bottom** parts shown inside the bubble, joined by `-`.

Example: an `LS` bubble (`LS / 1001`) on valve `THSITS-GSRSEL:PNE-VALVE-1002`
→ instrument tag `THSITS-GSRSEL:PNE-LS-1001`.
A `TT` bubble (`TT / 202`) on `THSITS-GSRSEL:PNE-PIPE-111`
→ `THSITS-GSRSEL:PNE-TT-202`.

ISA letter code (Sheet 2, subordinate to ISA-5.1 Table 4.1)

First letter = measured/initiating variable (+ optional modifier); succeeding letters = readout/passive, output/active, function modifier. The full table is in `legend_symbols.json` → `categories.instrumentation.letter_code`. Highlights:

`F` flow, `L` level, `P` pressure, `T` temperature, `A` analysis,
`S` speed/safety, `Z` position/SIS; succeeding `T` transmit, `I` indicate,
`C` control, `R` record, `S` switch, `V` valve/damper, `E` sensor/element.

In-line flow / primary elements (Sheet 3 — INSTRUMENT SYMBOLS)

`flow-sonic`, `flow-variable-area`, `flow-coriolis`, `flow-magnetic`,
`flow-vortex`, `flow-radiation`, `flow-orifice-generic`, `flow-nozzle`,
`flow-venturi`, `flow-positive-displacement`, `flow-turbine`.

3. Valves & actuators (Sheet 3)

The base body is the ISA two-triangle **bow-tie**. Lock/fail state codes are written above the body:

Code	Meaning	Code	Meaning
LC	Locked closed	NO	Open in normal operation
LO	Locked open	FC	Fail to closed (act-fail-closed)
NC	Closed in normal operation	FO	Fail to open (act-fail-open)
		FA	Fail as it is (act-fail-as-is)

Valve bodies: `valve-generic`, `valve-gate`, `valve-globe`, `valve-ball`,
`valve-needle`, `valve-butterfly`, `valve-angle`, `valve-three-way`,
`valve-check`, `valve-swing-check`, `valve-ball-check`, `valve-globe-check`,
`valve-tilting-disk-check`, `valve-piston-lift-check`,
`valve-y-piston-lift-check`, `valve-butterfly-check`, `valve-y-globe`,

valve-safety , valve-safety-spring-angle , valve-pressure-reducing ,
valve-balancing , valve-differential-pressure .

Actuating elements: act-diaphragm (pneumatic), act-piston ,
act-hydraulic , act-solenoid , act-manual , act-motor-electric ,
act-motor-dc , act-motor-pos-transmit , act-self-fcv/pcv/tcv/lcv
(self-acting control valves), act-fixed-spring , act-float , act-weight ,
act-high-speed , act-fail-closed/open/as-is .

4. Additional symbols

Equipment (Sheet 4)

Pumps (eq-pump-liquid , -centrifugal , -cavity), compressors/vacuum
(eq-compressor-vacuum , eq-turbo-compressor), eq-blower-fan , heat
exchangers (eq-heat-exchanger-general/-u-tube/-plate), eq-electric-heater ,
eq-steam-generator , eq-chiller , filters (eq-gas-filter , -hepa ,
eq-liquid-filter , eq-mixed-bed-filter , eq-ion-exchanger ,
eq-charcoal-filter), tanks (eq-horizontal-tank , -vertical-tank ,
-conical-tank , eq-hopper), eq-rupture-disc , eq-steam-trap ,
eq-viewing-glass , eq-orifice-plate-line .

Connection / end types (Sheet 3)

conn-threaded , conn-flanged-ends , conn-wafer , conn-welded-ends ,
conn-quick-coupling , conn-hose , conn-flanged-connection ,
conn-isolating-flange , conn-blind-flange , conn-screw-cap ,
conn-welded-cap , conn-reducer .

Interfaces & scope boundaries (Sheet 1)

- iface-termination-point — diamond TPXXXX / ZZZ . **X** = interface category: **B** building, **C** civil, **E** electrical, **G** compressed gasses, **H** HVAC, **L** liquid waste, **S** solid waste, **W** water; YYYY = unique number; ZZZ = next system/process.
- iface-offpage-connector — XXXXX | 123456-PID01 | WWWWW
(unique name | interconnecting P&ID number | from/to process; ZZZZZ = medium).
- iface-system-nfs — interface between (non-primary) systems, e.g. PBS1 .
- scope-design-limit — system & design-conditions limit (valve + diamond).
- scope-code-jurisdiction — piping **code jurisdiction break**
(e.g. ASME VIII ↔ ASME B31.3); * denotes the supplier side. Design conditions per code letter are tabulated in boxes on each drawing.

Miscellaneous / specialty (Sheet 5)

misc-reducer , misc-vfd , misc-flame-arrestor , misc-dielectric-joint ,
strainers (misc-strainer-y/-t/-cone), misc-sampling-point ,
misc-atmospheric-vent , misc-spray-nozzle , misc-agitator ,
misc-compensator , misc-funnel , misc-special-joint .

HVAC (Sheet 5)

hvac-shutoff-damper , hvac-parallel-damper , hvac-opposed-damper ,
 hvac-fan-general , hvac-radiator , hvac-heating-coil , hvac-cooling-coil ,
 hvac-filter , hvac-supply-terminal , hvac-exhaust-terminal .

Reference-only sets (Sheets 6-9, not redrawn)

These belong to facility-specific and electrical/fire single-line drawings, outside the core process-P&ID symbol scope. They are listed in

legend_symbols.json → reference_only_sets :

- **MINERVA RIB line specific elements** (Sheet 6) — pillow seal, clamped/thin/cooled windows, collimator, Hall sensor, magnet, target, bellows, etc.
 - **MINERVA beam optics** (Sheet 6) — beam position/halo/profile/tail monitors, quadrupole, wire scanner, steering plates, Faraday cup, electrostatic bender, mirrors, slits, beam shutter/dump, optical diagnostics, laser/ABU/FCU, etc.
 - **Electrical symbols** (Sheet 7) — IEC 60617; transformers, CTs, breakers, relays, contacts, meters, sounders, etc. (* = non-standard, ** = ANSI C37.2).
 - **Fire detection / extinguishing / safety** (Sheets 8-9) — detectors, sounders, extinguishing controls, fire separations E30/EI30/EI60/EI120, emergency / life-safety pictograms.
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Reconciliation with generator/symbols.py

The hand-built ISA-5.1 primitives in pid_project/generator/symbols.py remain valid but are **superseded by this authoritative legend** where they differ.

Recommended adoption path for the reconstruction:

<code>symbols.py</code> primitive	Authoritative legend equivalent	Note
<code>bubble()</code> (plain circle)	<code>inst-D1</code> (discrete, field)	field discrete instrument
<code>bubble_square()</code>	<code>inst-A1</code> (square + circle)	BPCS shared display
<code>dashed bubble (is_safety)</code>	<code>inst-B*</code> (SIS, square+circle+diagonals)	legend uses diagonals for SIS, not a dashed outline — align to <code>inst-B*</code>
<code>valve(kind="gate")</code>	<code>valve-gate</code>	bow-tie body matches
<code>valve(kind="control")</code>	<code>act-diaphragm</code>	diaphragm-actuated control valve
<code>valve(kind="solenoid")</code>	<code>act-solenoid</code>	matches
<code>valve(kind="relief")</code>	<code>valve-safety</code> / <code>valve-safety-spring-angle</code>	use angle type for spring-loaded relief
<code>vessel()</code>	<code>eq-vertical-tank</code> / <code>eq-horizontal-tank</code>	dished-end vessels
<code>heat_exchanger()</code>	<code>eq-heat-exchanger-general</code>	circle + zig-zag
<code>terminal_point()</code>	<code>iface-termination-point</code>	legend uses a diamond TPXXXX/ZZZ , not a crossed circle — switch to diamond

Key deviations to apply going forward

1. **SIS instruments** use a square+circle with **diagonal lines** (`inst-B*`), not a dashed outline.
2. **Termination points** are **diamonds** carrying `TPXXXX / ZZZ` , with the category letter set drawn from {B,C,E,G,H,L,S,W}.
3. **Line weights are normative**: primary = 1 mm, secondary = 0.5 mm, all signal lines = 0.25 mm. Future/proposed lines use dash-dot.
4. **Off-page continuation** uses the arrow-box connector with the interconnecting P&ID number (`123456-PID01`) and medium.
5. **Code jurisdiction breaks** are explicit symbols (valve + diamond + code letters) and must be placed wherever the piping code changes.

Regenerate everything with: `python3 standards/build_legend_library.py`